Power analysis via simulations

Readings for today

- Beaujean, A. A. (2014). Sample size determination for regression models using Monte Carlo methods in R. Practical Assessment, Research, and Evaluation, 19(1), 12.

Topics

1. Statistical Power

2. Monte Carlo methods

Statistical Power

Estimating data sufficiency

Q: Is my data sufficient to address my hypothesis? → If the effect is real, how likely am I to catch it?

Statistical Power: The ability to correctly reject a false H_0

n: sample size (certainty)

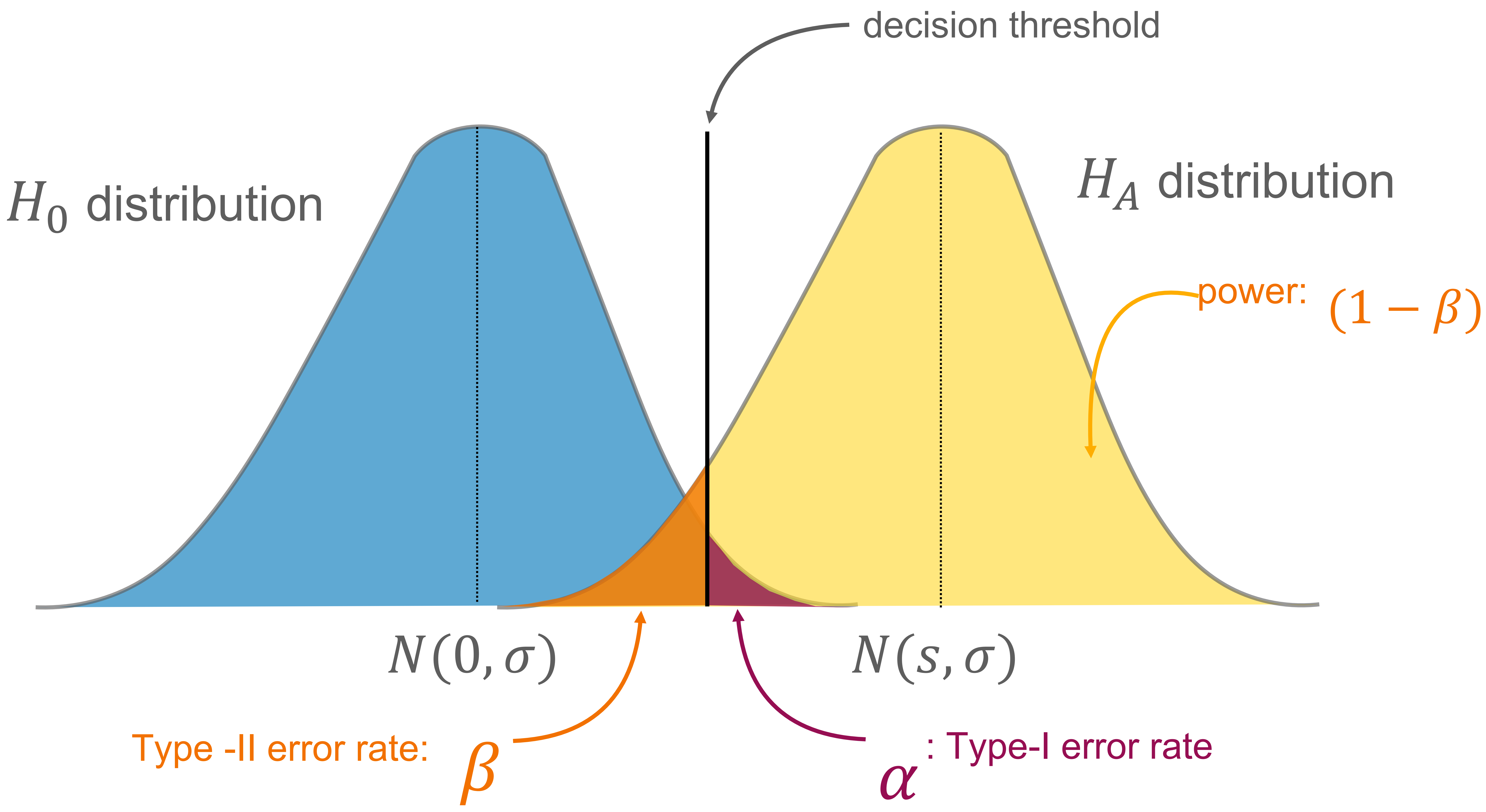
α: Type-I error (false positive) rate → p threshold

β: Type-II error (false negative) rate → Power = $1 - \beta$

σ: variance/noise: how messy the data is

s: effect size

Statistical power



(Beaujean 2014)

Goals of power analysis

Do I have enough data? → Do I have enough data to reject the null?

<u>Null Hypothesis Test Statistics (NHTS):</u> Do you have sufficient power to **detect** whether a test statistic is “statistically significant” (i.e., an effect)?

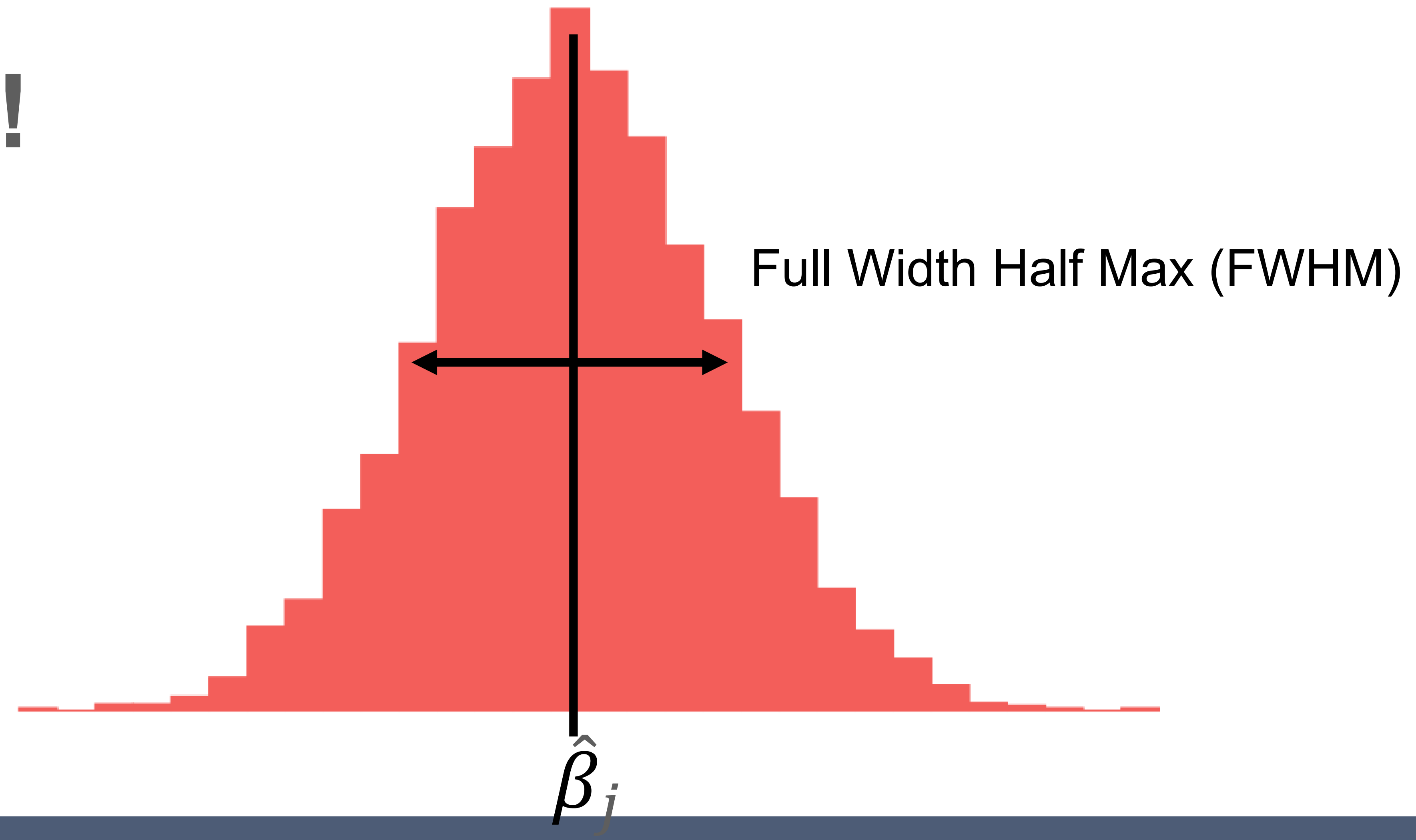
$P(reject\ H_0 | H_0\ is\ false)$

Power!

<u>Accuracy in Parameter Estimation (AIPE):</u> Do you have sufficient power to accurately recover the value of a statistical parameter (regardless of “statistical significance”)?

$MSE(\hat{\beta}_j - \beta_j)$

Confidence!



Do I have enough data to reproduce effects (t, β) that others have found?

(Beaujean 2014)

Traditional approaches to estimating power: detection

Goal: Sample size (n) determination. (e.g., Cohen's d)

Step 1: Search the literature for estimates of effect size (s). (e.g., Cohen's d)

Step 2: Determine best expectation of effective $E[s]$. (e.g., Literature says $d = 0.6$)

Step 3: Determine your ideal Type-I (α) and Type-II (β) error rates.

- I.e., define tolerance for being wrong (e.g., $\alpha = 0.05, \beta = 0.20$)

Step 4: Use *parametric* methods to calculate the sample size needed to achieve your α and β, assuming specific distributions of your data.

- $n = f(effect\ size, \sigma, \alpha, \beta)$

(Beaujean 2014)

Traditional approaches to estimating accuracy: precision

Goal: Parameter estimation accuracy ($\hat{s}$) determination.

Step 1: Determine the distributions of your X and Y. (e.g., normal, binary)

Step 2: Review the literature to estimate variability/uncertainty (e.g., variance,σ^2, SE)

Step 3: Determine your best expected model $E[\hat{f}(X)]$.

Step 4: Set desired confidence interval range for a given Type-I error rate (α)

Step 5: Use *parametric* methods to determine the sample size needed to achieve your target $(1-\alpha)\%$ confidence intervals.

- $n = f(desired\ CI\ width, \sigma, \alpha)$

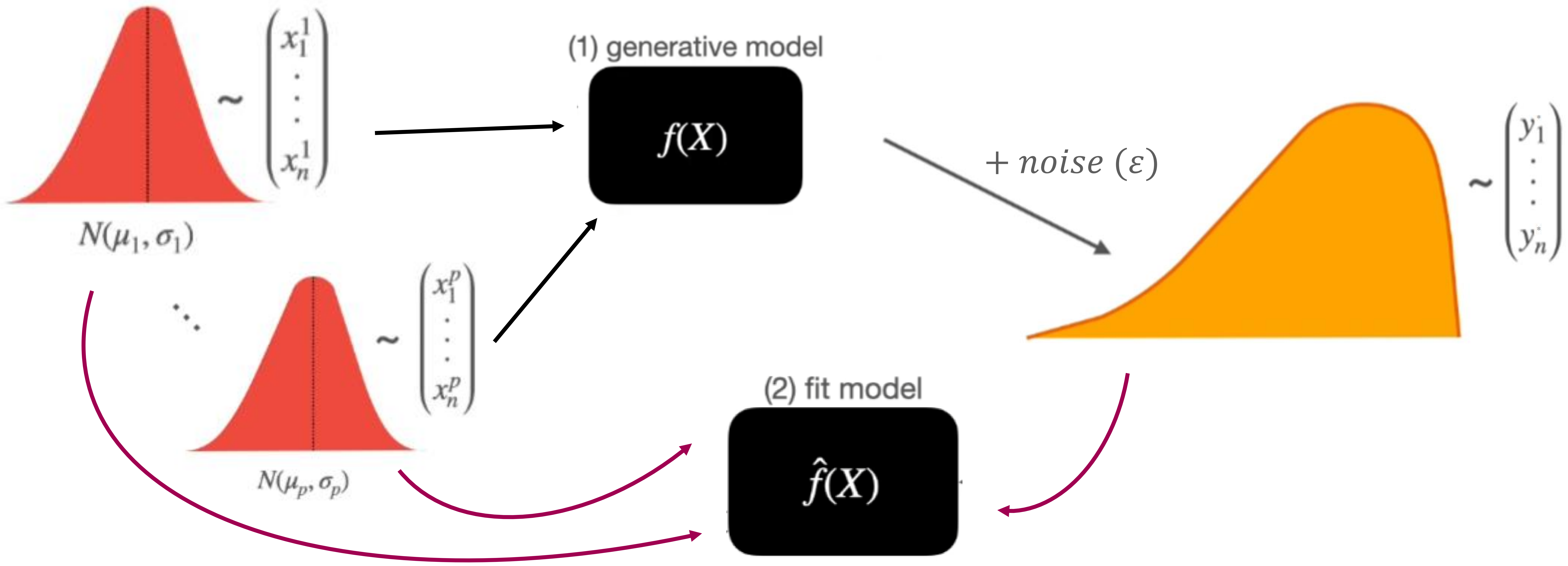
(Beaujean 2014)

Monte Carlo methods

Monte Carlo (MC) methods

Goal: Generate empirical distributions of your expected effects, instead of estimated distributions, to determine sufficiency of your data set.

MC method: Apply random sampling to simple data for numerical analysis



(Beaujean 2014)

MC power analysis

Step 1: Determine model of interest (e.g., regression, t-test)

Step 2: Decide on values for all parameters.

- Sample distributions: $X_p \sim N(\mu_p, \sigma_p)$ →
- Effect sizes: e.g., β_j

Distribution-based:
$Y^* \sim N(\mu_y, \sigma_y)$
Model-based:
$Y^* = \beta_0 + \beta_1 X + \varepsilon$
$\varepsilon \sim N(0, \sigma_\varepsilon)$

Step 3: Identify data "quirks" (e.g., missing value rate)

Step 4: Decide on aspects of MC simulations.

- α →Type-I error rate
- $(1-\beta)$ → power
- m → number of iterations/simulations
- n → sample size
- c → number of different random seeds (at least 2)
- σ_ε → residual variability in your outcome after accounting for your predictors

(Beaujean 2014)

MC power analysis

Step 5: Run m simulations using the parameters in Step 2.

Step 6: Evaluate performance of each simulation using quality metrics.

- If poor, $\uparrow n$ and repeat Step 5.

Step 7: Repeat Steps 5-6 using different random seeds. Check to see if results converge across seeds.

(Beaujean 2014)

MC power analysis algorithm

For $i = 1$ to m

i) Sample: e.g., $\underbrace{X_1 \sim N(\mu_1, \sigma_1)}_{normal}, \dots, \underbrace{X_j \sim B(n, p)}_{binomial}$

ii) Sim: $Y = f(x, \hat{\theta}) \rightarrow \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \boldsymbol{\epsilon}$

$\varepsilon \sim N(0, \sigma)$

$\sigma_\varepsilon = \sqrt{\left(\sum \beta_j^2\right) \cdot \frac{1 - R^2}{R^2}}$

iii) Evaluate: $T(y, x) \rightarrow \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$

iv) Track: $T = [T_1(Y_1, X_1), \dots, T_i(Y_i, X_i)]$ also, count # of times $p < \alpha$

(Beaujean 2014)

MC simulation quality metrics

Relative parameter estimate bias:

$$\theta_{bias} = \frac{\hat{\theta} - \theta_H}{\theta_H}$$

- θ_H: pre-set (real) effect sizes
- θ : average fit parameter across simulations

Relative standard error bias:

$$\sigma_{bias} = \frac{\hat{\sigma}_\theta - \sigma_{\hat{\theta}}}{\sigma_{\hat{\theta}}}$$

- $\hat{\sigma}_\theta$: average fit of parameter SE across simulations
- $\sigma_{\hat{\theta}}$: standard deviation of $\hat{\theta}$ across simulations

Coverage:

$$c = \frac{1}{m} I(\theta \neq 0)$$

$$I(\theta \neq 0) = \begin{cases} 1, \text{ if } H_0 = \text{rejected} \\ 0, \text{ otherwise} \end{cases}$$

(Beaujean 2014)

Example: Regression

Model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

$$Y^* \sim N(\beta_0 + \beta_1 X, \sigma_\epsilon)$$

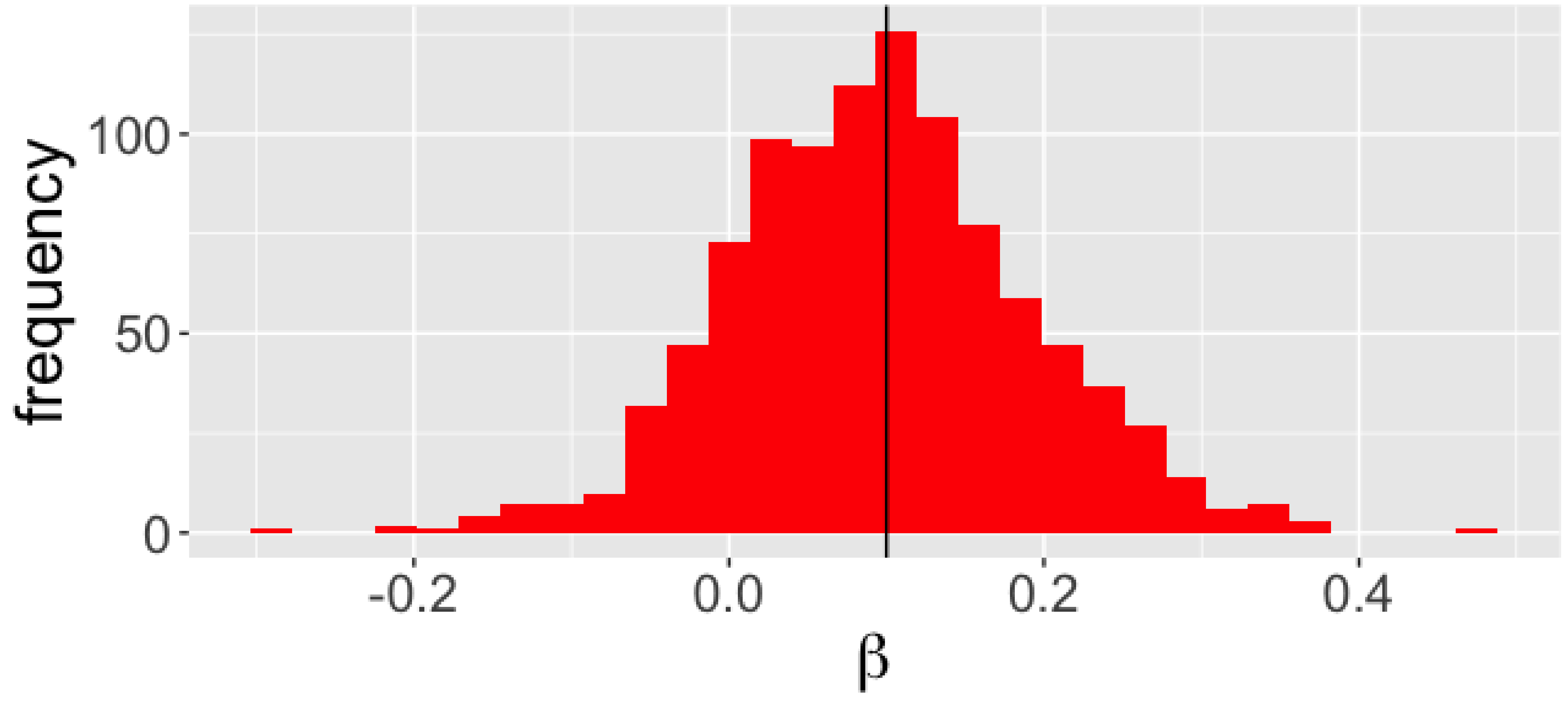
Parameters:

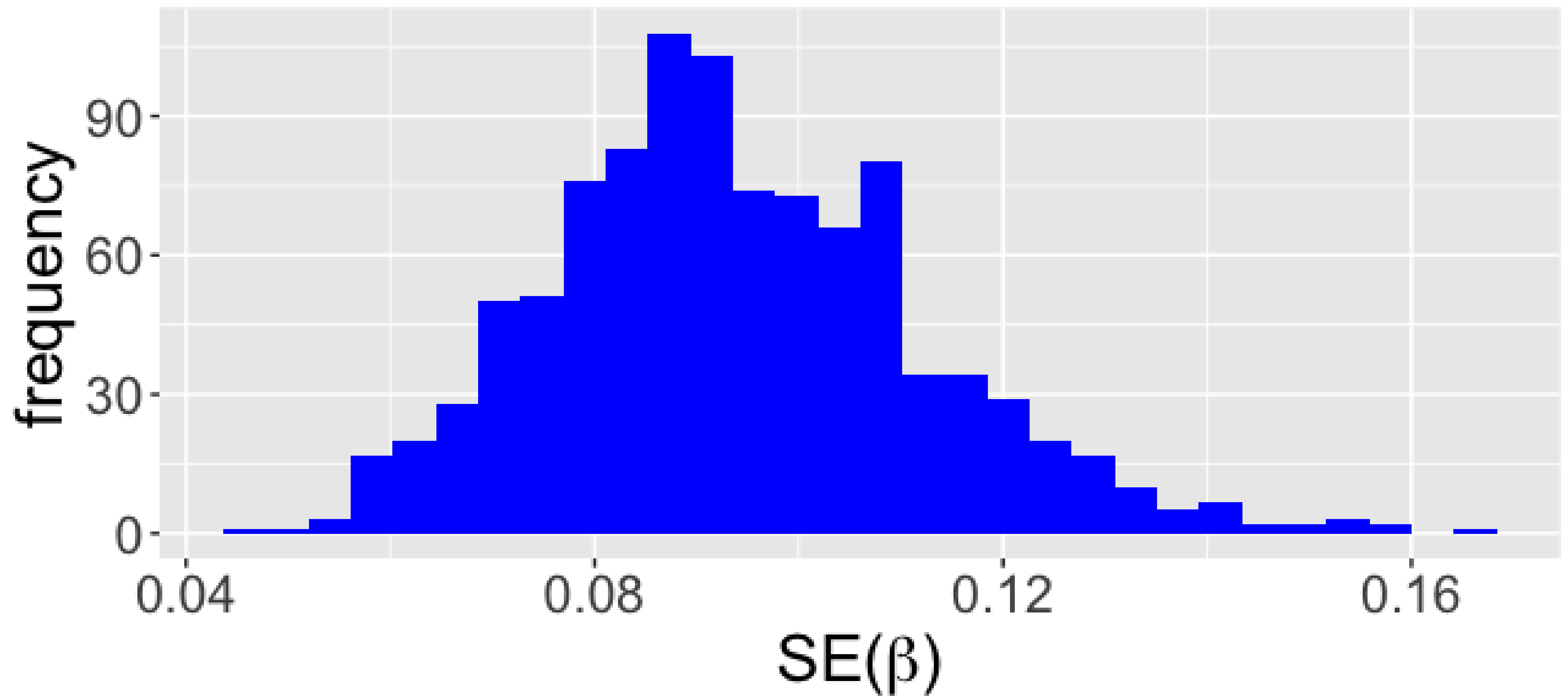
$\alpha = 0.05$ $m = 1000$

$X \sim N(3,2)$ $n = 30$

$\beta_0 = 9$ $\beta_1 = 0.1$

$\epsilon \sim N(0,1) = N(0, \sigma_\epsilon)$





Example: Regression

<u>Relative parameter estimate bias:</u>

$$\theta_{bias} = \frac{\hat{\theta} - \theta_H}{\theta_H} = -0.045$$

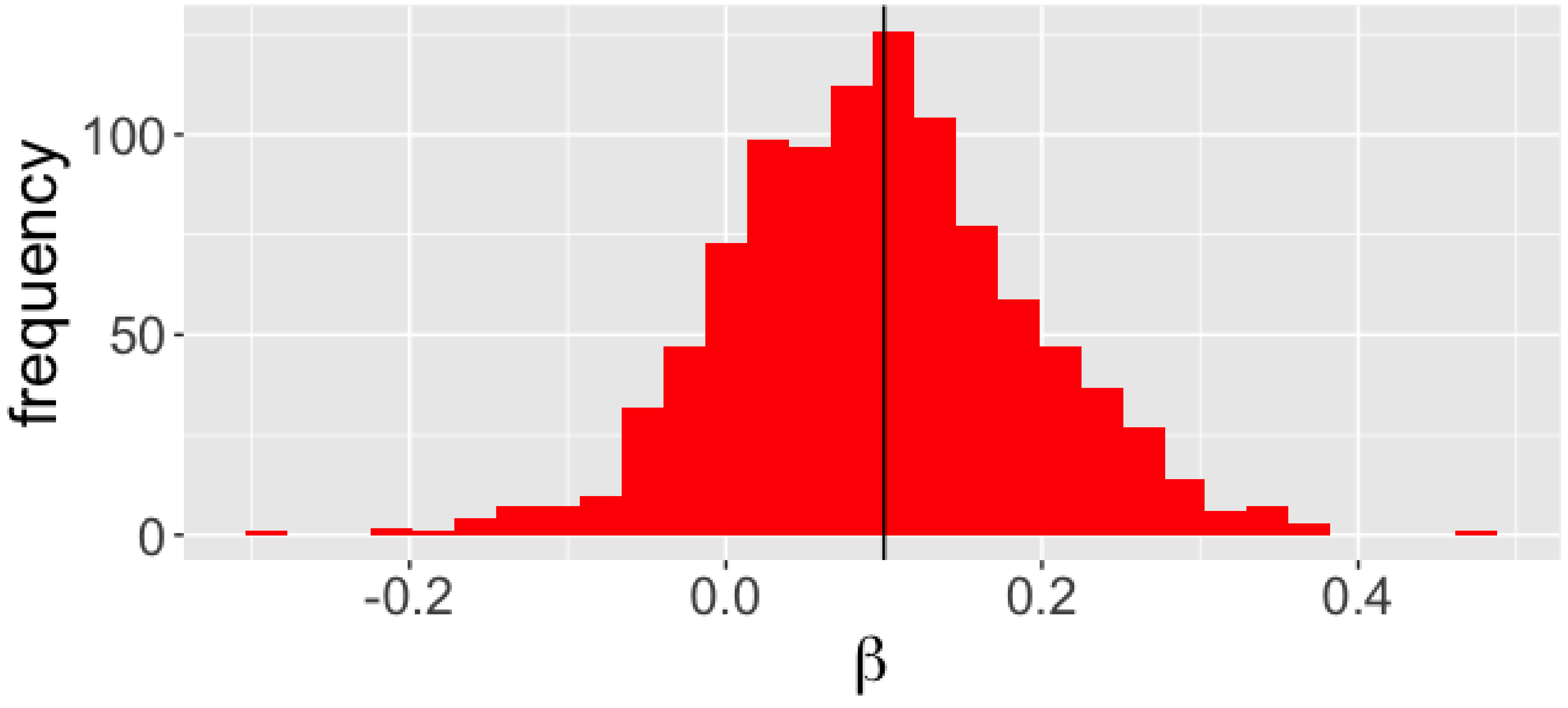
<u>Relative standard error bias:</u>

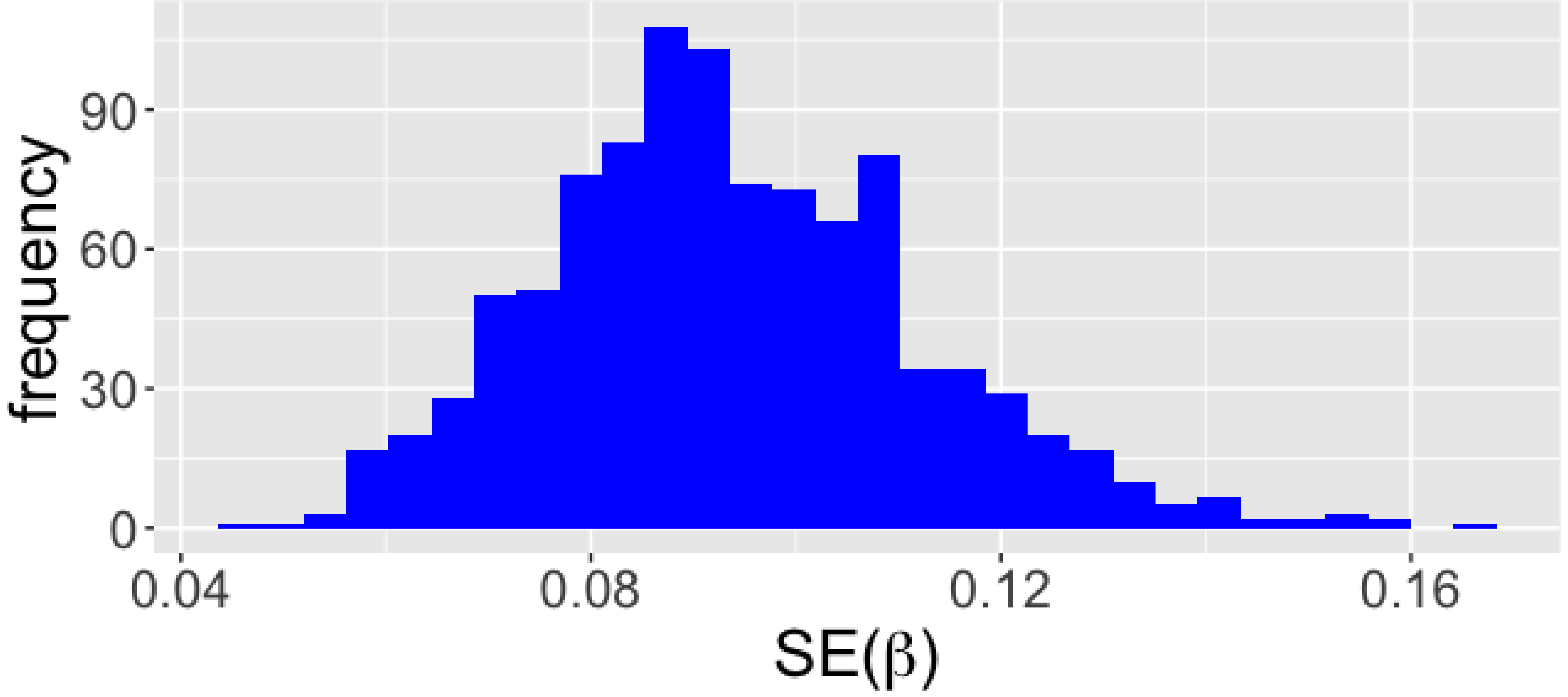
$$\sigma_{bias} = \frac{\hat{\sigma}_{\theta} - \sigma_{\hat{\theta}}}{\sigma_{\hat{\theta}}} = -0.010$$

<u>Coverage:</u>

$$c = \frac{1}{m} I(\theta \neq 0) = 0.173$$

how often your confidence intervals contains the true value

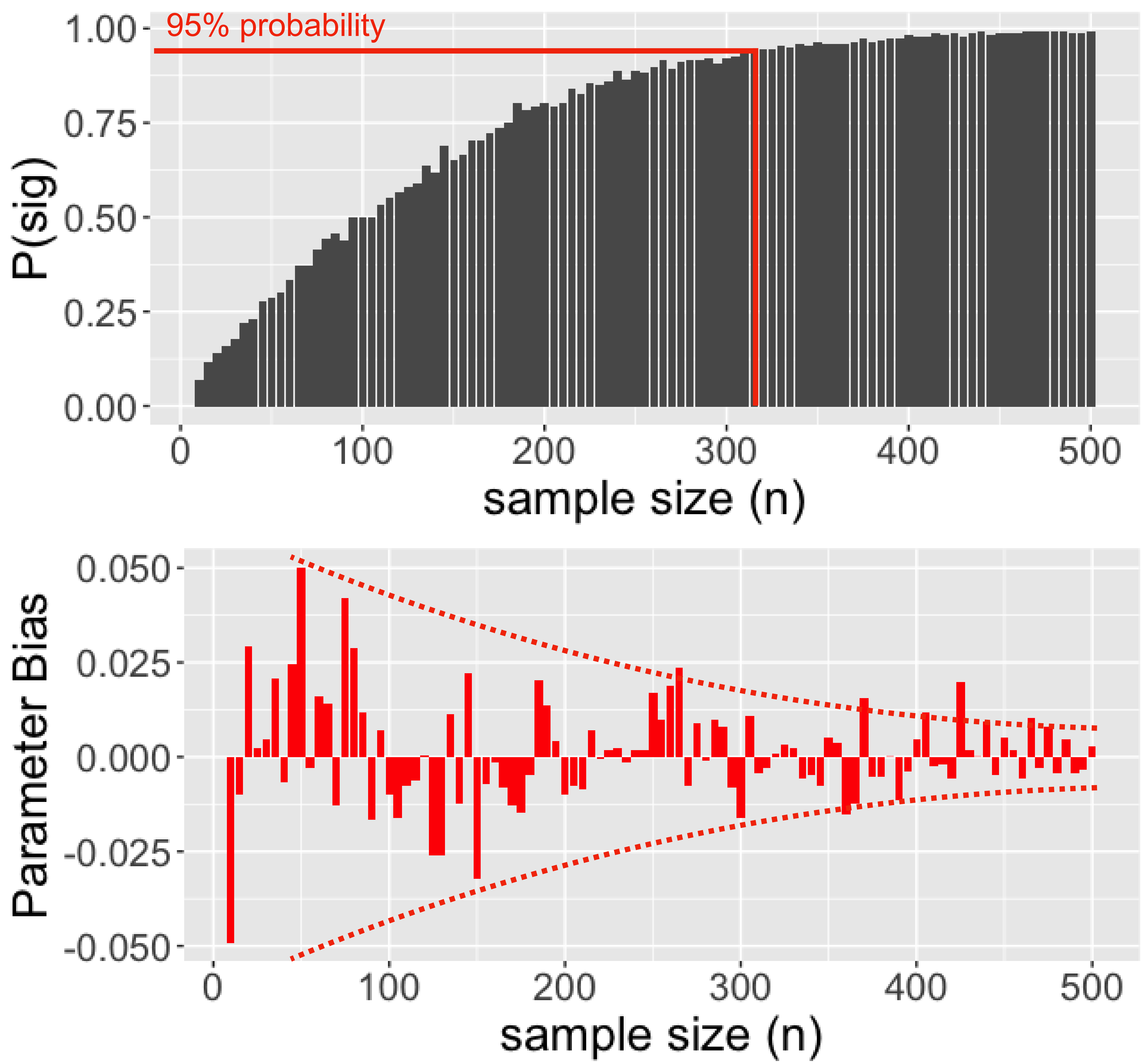




Example: power & accuracy

Goal: Determine the sample size (n) needed in order to:

1. Detect a significant effect of β_1 95% of the time. (Coverage)
2. Have a parameter bias $< |0.01|$



Approach: Repeat model sims from before, across a range of sample sizes (n).

$$n = \underbrace{10 - 500}_{range}$$

Take home message

- Because they rely on relatively few assumptions, Monte Carlo methods provide a robust and flexible way of estimating the statistical power of your model.